5.1

$t_{ox} := 4\ nm$ $\mu_n := 450\ \dfrac{cm^2}{V \cdot s}$ $v_t := 0.5\ V$ $\varepsilon_{ox} := 3.9\ \varepsilon_0$

$$C_{ox} := \frac{\varepsilon_{ox}}{t_{ox}}$$

$$k'_n := \mu_n \cdot C_{ox} = 388.477\ \frac{\mu A}{V^2}$$

$r_{DS} := 1\ k\Omega$ $v_{GS} := 1\ V$ $L := 0.18\ \mu m$

$$W := \frac{L}{\mu_n \cdot C_{ox} \cdot (v_{GS} - v_t) \cdot r_{DS}} = 0.927\ \mu m$$

5.2

$t_{ox} := 1.4\ nm$ $L := 65\ nm$ $\mu_n := 216\ \dfrac{cm^2}{V \cdot s}$ $v_t := 0.35\ V$

$C_{ox} := \dfrac{\varepsilon_{ox}}{t_{ox}} = 0.025\ \dfrac{pF}{\mu m^2}$ $k'_n := \mu_n \cdot C_{ox} = 532.769\ \dfrac{\mu A}{V^2}$ $W := 10 \cdot L$

$i_D := 50\ \mu A$

$v_{OV} := \sqrt{\dfrac{i_D}{0.5 \cdot k'_n \cdot 10}} = 0.137\ V$ $v_{GS} := v_{OV} + v_t = 0.487\ V$

$v_{DS\_min} := v_{OV} = 0.137\ V$

5.6

$L := 0.8\ \mu m$ $k'_n := 400\ \dfrac{\mu A}{V^2}$ $V'_A := 5\ \dfrac{V}{\mu m}$ $W := 16\ \mu m$

$V_A := V'_A \cdot L = 4\ V$ $\lambda := \dfrac{1}{V_A} = 0.25\ \dfrac{1}{V}$

$v_{OV} := 0.2\ V$ $v_{DS} := 0.8\ V$

$i_D := \dfrac{1}{2} \cdot k'_n \cdot \dfrac{W}{L} \cdot v_{OV}^2 \cdot (1 + \lambda \cdot v_{DS}) = 192\ \mu A$ $I'_D := \dfrac{1}{2} \cdot k'_n \cdot \dfrac{W}{L} \cdot v_{OV}^2$

$r_o := \dfrac{V_A}{I'_D} = 25\ k\Omega$

$i_{D2} := \dfrac{1}{2} \cdot k'_n \cdot \dfrac{W}{L} \cdot v_{OV}^2 \cdot (1 + \lambda \cdot (v_{DS} + 1\ V)) = 232\ \mu A$

$i_{D2} - i_D = 40\ \mu A$